Talagrand's discrete-convexity conjecture

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

We prove Talagrand's discrete-convexity conjecture. There is a universal integer k such that, whenever an arbitrary family has Bernoulli product measure at least $1 - 1/k$, the sets not contained in a union of k members admit a cover of total cost at most $1/2$ at the same density.

1 The result and how it is proved

1.1 Setting

Throughout, $N \geq 1$ is an integer, $[N] = \{1, \ldots, N\}$, and a *family* is a subset of $2^{[N]}$. For $p \in (0, 1)$, μ_p is the probability measure on $2^{[N]}$ under which each element of $[N]$ is present independently with probability p; that is, $\mu_p(\{S\}) = p^{|S|}(1-p)^{N-|S|}$.

Exceptional sets. For a family $\mathcal{D}$ and an integer $k \geq 1$, the family $E_k(\mathcal{D})$ consists of the sets that *cannot* be covered by k members of $\mathcal{D}$:

$$S \in E_k(\mathcal{D}) \iff \text{there is no tuple } (D_1, \ldots, D_k) \in \mathcal{D}^k \text{ with } S \subseteq D_1 \cup \cdots \cup D_k. \tag{1.1}$$

The members of a tuple may repeat and need not be disjoint. Two facts follow at once from the definition and are used later.

- If $\mathcal{D} \neq \varnothing$, then $\varnothing \notin E_k(\mathcal{D})$, because $\varnothing \subseteq D \cup \cdots \cup D$ for any $D \in \mathcal{D}$.
- $E_k(\mathcal{D})$ is increasing: if $S \in E_k(\mathcal{D})$ and $S \subseteq S'$, then $S' \in E_k(\mathcal{D})$, because a union containing S' would contain S.

Covers and smallness. A family $\mathcal{G}$ *covers* a family $\mathcal{A}$ if every $S \in \mathcal{A}$ contains some $I \in \mathcal{G}$. The members of $\mathcal{G}$ are called *generators*. For $\rho \in (0, 1)$ the *cost of* $\mathcal{G}$ *at density* ρ is

$$c_\rho(\mathcal{G}) = \sum_{I \in \mathcal{G}} \rho^{|I|}.$$

A family $\mathcal{A}$ is ρ-*small* if it has a cover $\mathcal{G}$ with $c_\rho(\mathcal{G}) \leq 1/2$. Three remarks:

- Since $\mu_p\{S : I \subseteq S\} = p^{|I|}$, a cover $\mathcal{G}$ of $\mathcal{A}$ gives $\mu_p(\mathcal{A}) \leq c_p(\mathcal{G})$ by a union bound. So p-smallness is an explicit certificate that $\mu_p(\mathcal{A}) \leq 1/2$.
- $c_\rho(\mathcal{G})$ is nondecreasing in ρ. Hence ρ-small implies ρ'-small for every $\rho' \leq \rho$: smallness at a *lower* density is a *weaker* statement.
- The empty family $\mathcal{A} = \varnothing$ is covered by $\mathcal{G} = \varnothing$ at cost 0.

A toy example. Let $N = 3$, $\mathcal{D} = \{\{1\}, \{2\}\}$, $k = 2$. The unions of two members are $\{1\}, \{2\}, \{1, 2\}$, so $E_2(\mathcal{D}) = \{S : 3 \in S\}$. It is covered by $\mathcal{G} = \{\{3\}\}$, whose cost at density p is p.

1.2 The main theorem

Talagrand formulated the discrete-convexity conjecture in [11, Conjecture 7.1]; the same formulation, at the original density p, appears as Research Problem 13.3.2 in [12]. In the notation above, and as the original manuscript states it, the conjecture is: *there is a universal integer k such that $\mu_p(\mathcal{D}) \geq 1 - 1/k$ implies that $E_k(\mathcal{D})$ is p-small.* Informally, if $\mathcal{D}$ is sufficiently likely, then every set outside a p-small family is contained in a union of k members of $\mathcal{D}$. The following theorem proves it with $k = 2^{75}$.

Theorem 1.1. *Set $k = 2^{75}$. For every integer $N \geq 1$, every $p \in (0, 1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) \geq 1 - \frac{1}{k} \quad \Longrightarrow \quad E_k(\mathcal{D}) \text{ is } p\text{-small.}$$

The constant is independent of N, p, and $\mathcal{D}$. The family $\mathcal{D}$ is arbitrary; no monotonicity is assumed. The measure in the hypothesis and the density in the conclusion are the same p.

1.3 Proof at a glance

Key idea. Fix a family $\mathcal{F}$ and a density q with $\mu_q(\mathcal{F}) \geq 1/2$. To every $U \subseteq [N]$ we attach a weight $w(U) \geq 0$ (a rescaled squared Fourier coefficient of the indicator of $\mathcal{F}$) with two properties.

(W1) *Globally the weights are small:* $\sum_{U \subseteq [N]} q^{|U|} w(U) \leq 1$. This follows from Parseval's identity once w is expressed through Fourier coefficients (Proposition 3.1, Step 1).

(W2) *On every exceptional set they are large:* every $S \in E_{32}(\mathcal{F})$ satisfies $\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \geq 2^{-32}$. This comes from a signed measure on $32 \times N$ arrays that vanishes unless the 32 rows jointly cover S (Proposition 3.1, Step 2).

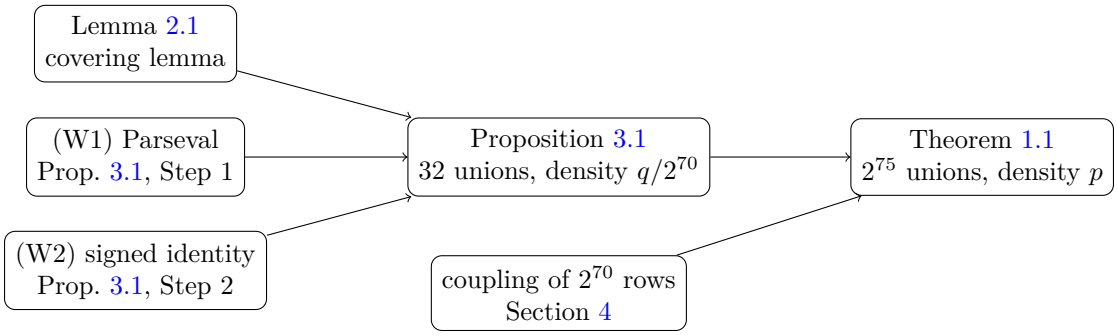
By (W2), an exceptional set must contain either one subset U of large weight, or very many subsets of the same size and comparable smaller weight. By (W1), the subsets of large weight are cheap to use as generators, and each class of "same size, comparable weight" has bounded total cost. A separate combinatorial lemma covers, cheaply, all sets containing very many members of such a class.

The steps and their dependencies.

1. **Lemma 2.1** (Section 2), a self-contained covering lemma. Let r, h be integers with $1 \leq r \leq h$ and $h \geq 64$. If an r-uniform family $\mathcal{H}$ has $|\mathcal{H}|\rho^r \leq 2^{h+1}$, then the sets containing at least 2^{12h} members of $\mathcal{H}$ have a cover of cost at most $3 \cdot 2^{-h}$ at density ρ. The proof is a random construction.

2. **Proposition 3.1** (Section 3). If $\mu_q(\mathcal{F}) \geq 1/2$, then $E_{32}(\mathcal{F})$ is $(q/2^{70})$-small. It proves (W1) and (W2), takes the heavy sets as generators, sorts the others into bins by size r and by dyadic weight scale 2^{-h}, and applies Lemma 2.1 to each bin. A set that avoids all generators contains fewer than 2^{12h} sets of each bin, each contributing at most 2^{-16h} to the sum in (W2), so the sum is below 2^{-32} and the set is not exceptional. The sixteenth power is what makes $2^{12h} \cdot 2^{-16h} = 2^{-4h}$ summable.

3. **Proof of Theorem 1.1** (Section 4). Step 2 loses a factor $L = 2^{70}$ in the density. To undo this, take L random sets $R_1, \dots, R_L$, each with law exactly μ_p, coupled as follows: independently across coordinates, each coordinate is placed in at most one of the sets, and it is placed in some set with probability Lp. The sets are then pairwise disjoint, and because the choices at different coordinates are independent, their union has law μ_{Lp}. Apply step 2 with $q = Lp$ to the family $\mathcal{F}$ of unions of L members of $\mathcal{D}$. Then $E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D})$ and $q/L = p$. This needs $Lp < 1$; when $Lp \geq 1$ a different coupling shows directly that $E_k(\mathcal{D}) = \varnothing$.



Where the constants come from. $k = 32 \cdot 2^{70}$. The 32 is the number of rows in (W2), giving the exponent $32 = 2 \cdot 16$. The factor $2^{70} = 2^{64} \cdot 2^6$ arises in Proposition 3.1: the threshold $2^{-64|U|}$ for "heavy" sets guarantees that every bin has $h \geq 64$, as Lemma 2.1 requires, and the extra 2^{-6} makes the heavy generators cost at most $1/64$.

What is not on the path to Theorem 1.1. Section 5 collects a consequence for positive selector processes, a two-union corollary that depends on external results, and pointers to related work. None of it is used in Sections 2–4.

Notation changes relative to the original manuscript. To keep one meaning per symbol, a few letters are renamed. The signed column mass in Section 3 is $\widetilde{P}$ (originally R, which clashes with the random rows R_j). The random point of the cube in Section 3 is ξ (originally X, which clashes with the ground set of Lemma 2.1). The inclusion probability in Case 2 of Section 4 is τ (originally t). In Section 5 the selector class is $\mathcal{T}$ with elements θ (originally T and t), and the density in the two-union corollary is σ (originally r). In Section 2, π is a probability, not the circle constant, and e, e' are edges, so the exponential is written exp.

2 An elementary covering lemma

Where this is used. Only in Step 4 of the proof of Proposition 3.1, with $X = [N]$, $\rho = q/2^{70}$, and $\mathcal{H}$ a bin of sets of size r whose weights are about 2^{-h}.

What it says. Think of ρ^r as the cost of one member ("edge") of an r-uniform family $\mathcal{H}$. The total cost $|\mathcal{H}|\rho^r$ may be as large as 2^{h+1}, so the edges themselves are far too expensive to serve as generators. The lemma says that the sets containing a huge number, 2^{12h}, of edges can nevertheless be covered at tiny cost. Unions of sampled edges also appear in the covering constructions of Fischer and Person [3, Sections 1.1 and 2]; the argument below is self-contained.

Lemma 2.1. *Let X be a finite set, let $0 < \rho < 1$, and let r, h be integers with $1 \le r \le h$ and $h \ge 64$. Suppose that $\mathcal{H} \subseteq \binom{X}{r}$ satisfies*

$$|\mathcal{H}|\rho^r \le 2^{h+1}.$$

Then the family

$$\Big\{S \subseteq X : |\{e \in \mathcal{H} : e \subseteq S\}| \ge 2^{12h}\Big\}$$

has a cover $\mathcal{G} \subseteq 2^X$ with

$$c_\rho(\mathcal{G}) \le 3 \cdot 2^{-h}.$$

Proof. Plan. We build $\mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2$. The part $\mathcal{C}_0$ is deterministic; $\mathcal{C}_1$ and $\mathcal{C}_2$ depend on a random sample. We show that $\mathcal{G}$ is a cover for *every* outcome of the sample (Step 4) and that its *expected* cost is at most $3 \cdot 2^{-h}$ (Steps 1, 3, 5). Some outcome then has cost at most the expectation (Step 6).

Call the family in the statement the *target family*. Put

$$m = 2^{12h}, \qquad T = 2^{4h}, \qquad \pi = 2^{-8h}, \qquad \text{so that} \qquad \pi m = 2^{4h}.$$

Step 1 (sets of large weighted degree: the generators $\mathcal{C}_0$)**.** For $I \subseteq X$ with $|I| \le r$, define the weighted degree

$$d(I) = |\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|}.$$

Thus $d(I)\rho^{|I|}$ is the total cost ρ^r of the edges containing I, and $d(I) > T$ means that I is more than T times cheaper than the edges through it. Let

$$\mathcal{C}_0 = \{I \subseteq X : |I| \le r,\ d(I) > T\}.$$

Exchanging the order of summation, and using that each edge has exactly 2^r subsets,

$$\sum_{\substack{I \subseteq X \\ |I| \le r}} d(I)\rho^{|I|} = \sum_I \sum_{\substack{e \in \mathcal{H} \\ e \supseteq I}} \rho^r = \rho^r \sum_{e \in \mathcal{H}} |\{I : I \subseteq e\}| = 2^r |\mathcal{H}| \rho^r.$$

For $I \in \mathcal{C}_0$ we have $1 < d(I)/T$. Therefore

$$c_\rho(\mathcal{C}_0) = \sum_{I \in \mathcal{C}_0} \rho^{|I|} \le \frac{1}{T} \sum_{I \in \mathcal{C}_0} d(I)\rho^{|I|} \le \frac{2^r |\mathcal{H}| \rho^r}{T} \le 2^{-4h} \cdot 2^r \cdot 2^{h+1} = 2^{r+1-3h} \le 2^{1-2h}. \tag{2.1}$$

The third inequality is the hypothesis $|\mathcal{H}|\rho^r \le 2^{h+1}$; the last uses $r \le h$.

Step 2 (regular edges and the extension inequality)**.** Call an edge $e \in \mathcal{H}$ *regular* if it contains no member of $\mathcal{C}_0$, and let $\mathcal{H}'$ be the family of regular edges. Then

$$I \subseteq e \ \text{ for some } e \in \mathcal{H}' \quad \Longrightarrow \quad d(I) \le T. \tag{2.2}$$

Indeed, such an I has $|I| \le r$, and $I \notin \mathcal{C}_0$ because the regular edge e contains no member of $\mathcal{C}_0$. By the definition of $\mathcal{C}_0$, a set of size at most r outside $\mathcal{C}_0$ has $d(I) \le T$.

Claim. For every $B \subseteq X$,

$$\sum_{e \in \mathcal{H}'} \rho^{|e \setminus B|} \le T \sum_{i=0}^{r} \binom{|B|}{i}. \tag{2.3}$$

Proof of the claim. Sort the regular edges e according to the set $I = e \cap B$. This I is a subset of B of size at most r, so there are at most $\sum_{i=0}^{r} \binom{|B|}{i}$ possible values of I. Fix an I for which the group $\{e \in \mathcal{H}' : e \cap B = I\}$ is nonempty. Every e in the group has $|e \setminus B| = r - |I|$ and contains I, so

$$\sum_{\substack{e \in \mathcal{H}' \\ e \cap B = I}} \rho^{|e \setminus B|} = |\{e \in \mathcal{H}' : e \cap B = I\}|\, \rho^{r-|I|} \le |\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|} = d(I) \le T.$$

The last inequality is (2.2), which applies because the group is nonempty. Summing over the possible I gives (2.3).

Step 3 (sampled pairs: the generators $\mathcal{C}_1$)**.** Independently for each ordered pair $(e, e') \in (\mathcal{H}')^2$, including the diagonal pairs (e, e), *sample* that pair with probability π. Let $\mathcal{C}_1$ be the set of unions $e \cup e'$ of sampled pairs.

Each member of $\mathcal{C}_1$ is the union of at least one sampled pair, so $c_\rho(\mathcal{C}_1) \le \sum_{(e,e') \text{ sampled}} \rho^{|e \cup e'|}$. Here a union arising from several sampled pairs is counted several times on the right. Taking expectations (linearity only; independence is not used here),

$$\begin{aligned} \mathbb{E}[c_\rho(\mathcal{C}_1)] &\le \pi \sum_{(e,e') \in (\mathcal{H}')^2} \rho^{|e \cup e'|} = \pi \sum_{e \in \mathcal{H}'} \rho^r \sum_{e' \in \mathcal{H}'} \rho^{|e' \setminus e|} \\ &\le \pi\, |\mathcal{H}'|\, \rho^r\, 2^r T \le 2^{-8h} \cdot 2^{h+1} \cdot 2^r \cdot 2^{4h} = 2^{r+1-3h} \le 2^{1-2h}. \end{aligned} \tag{2.4}$$

The equality uses $|e \cup e'| = |e| + |e' \setminus e| = r + |e' \setminus e|$. The next inequality is (2.3) with $B = e$, where $|B| = r$ and $\sum_{i=0}^{r} \binom{r}{i} = 2^r$. Then we use $|\mathcal{H}'|\rho^r \le |\mathcal{H}|\rho^r \le 2^{h+1}$ and finally $r \le h$.

Step 4 (residual tuples: the generators $\mathcal{C}_2$; $\mathcal{G}$ is always a cover)**.** For every ordered tuple $(e_1, \dots, e_m)$ of *distinct* regular edges, put $e_1 \cup \cdots \cup e_m$ into $\mathcal{C}_2$ if none of the m^2 pairs (e_i, e_j), $1 \le i, j \le m$ (including $i = j$), was sampled. Set

$$\mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2.$$

We check that $\mathcal{G}$ covers the target family for every sampling outcome. Let S be a target set. If S contains a member of $\mathcal{C}_0$, we are done. Otherwise every edge $e \in \mathcal{H}$ with $e \subseteq S$ is regular: a member of $\mathcal{C}_0$ inside e would lie inside S. Since S is a target set, it contains at least m edges, all regular; choose m distinct ones and order them as $(e_1, \dots, e_m)$. If some pair (e_i, e_j) was sampled, then $e_i \cup e_j \in \mathcal{C}_1$ and $e_i \cup e_j \subseteq S$. If no such pair was sampled, then $e_1 \cup \cdots \cup e_m \in \mathcal{C}_2$ and it is contained in S.

Step 5 (expected cost of $\mathcal{C}_2$)**.**

5a. Total weight of tuples. For $0 \le t \le m$, let

$$W_t = \sum_{\substack{(e_1,\dots,e_t) \in (\mathcal{H}')^t \\ e_1,\dots,e_t \text{ distinct}}} \rho^{|e_1 \cup \cdots \cup e_t|}, \qquad W_0 = 1$$

(for $t = 0$ there is one empty tuple, with empty union). We claim

$$W_{t+1} \le T(1 + rm)^r\, W_t \quad (0 \le t < m), \qquad \text{hence} \qquad W_m \le \big[T(1 + rm)^r\big]^m. \tag{2.5}$$

Fix a tuple $(e_1, \dots, e_t)$ of distinct regular edges with $t < m$ and let $B = e_1 \cup \cdots \cup e_t$, so $|B| \le rt \le rm$. First,

$$\sum_{i=0}^{r} \binom{|B|}{i} \le (1 + |B|)^r \le (1 + rm)^r.$$

To see the first inequality, fix a linear order on B and a new symbol $*$. Map a subset $\{b_1 < \cdots < b_i\}$ of B with $i \le r$ to the sequence $(b_1, \dots, b_i, *, \dots, *)$ of length r; this is an injection into $(B \cup \{*\})^r$. Second, for any edge e we have $\rho^{|B \cup e|} = \rho^{|B|}\rho^{|e \setminus B|}$. Hence, by (2.3),

$$\sum_{e \in \mathcal{H}' \setminus \{e_1, \dots, e_t\}} \rho^{|B \cup e|} \le \rho^{|B|} \sum_{e \in \mathcal{H}'} \rho^{|e \setminus B|} \le \rho^{|B|}\, T(1 + rm)^r.$$

(Allowing all regular edges e, not only the new ones, can only increase the sum.) The left side is the total weight of the $(t+1)$-tuples extending $(e_1, \dots, e_t)$, and $\rho^{|B|}$ is the weight of that t-tuple in W_t. Summing over all tuples $(e_1, \dots, e_t)$ gives the recursion in (2.5). Iterating from $W_0 = 1$ gives the bound on W_m.

5b. Survival probability of one tuple. Fix a tuple $(e_1, \dots, e_m)$ of distinct regular edges. Because the e_i are distinct, the m^2 ordered pairs (e_i, e_j) are m^2 *different* elements of $(\mathcal{H}')^2$. These pairs are sampled independently, so the probability that none of them is sampled is exactly

$$(1 - \pi)^{m^2} \le \exp(-\pi m^2) \le 2^{-\pi m^2}.$$

Here $1 - \pi \le \exp(-\pi)$ is the inequality $1 + u \le \exp(u)$, and $\exp(-x) = 2^{-x/\ln 2} \le 2^{-x}$ for $x \ge 0$ because $\ln 2 < 1$ gives $x/\ln 2 \ge x$. This is the only place where independence of the sample is used.

5c. Expected cost. Each member of $\mathcal{C}_2$ is the union of at least one tuple none of whose pairs was sampled. Counting with multiplicity as in Step 3 and using linearity of expectation together with 5a and 5b,

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le W_m\,(1 - \pi)^{m^2} \le \left[T(1 + rm)^r\right]^m 2^{-\pi m^2}.$$

We now show that the right side is at most 2^{-m}. Using $r \le h$ and $h + 1 \le 2^h$,

$$1 + rm \le 1 + h\,2^{12h} \le (h+1)2^{12h} \le 2^{13h}, \qquad \text{so} \qquad \log_2\left[T(1 + rm)^r\right] \le 4h + 13rh.$$

Next, using $r \le h$, then $4h + 1 \le 5h \le 5h^2$, then $h \le 2^h$, and finally $18 \le 2^{2h}$ (true for $h \ge 3$, and we have $h \ge 64$),

$$4h + 13rh + 1 \le 4h + 13h^2 + 1 \le 18h^2 \le 18 \cdot 2^{2h} \le 2^{4h} = \pi m.$$

Multiplying $4h + 13rh \le \pi m - 1$ by m gives $(4h + 13rh)m - \pi m^2 \le -m$. It follows that

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le 2^{(4h+13rh)m - \pi m^2} \le 2^{-m}. \tag{2.6}$$

Step 6 (conclusion)**.** Since $c_\rho(\mathcal{G}) \le c_\rho(\mathcal{C}_0) + c_\rho(\mathcal{C}_1) + c_\rho(\mathcal{C}_2)$, the bounds (2.1), (2.4), and (2.6) give

$$\mathbb{E}[c_\rho(\mathcal{G})] \le 2^{1-2h} + 2^{1-2h} + 2^{-m} = 2^{2-2h} + 2^{-m} \le 2^{-h} + 2^{-h} \le 3 \cdot 2^{-h}.$$

Here $2^{2-2h} \le 2^{-h}$ because $h \ge 2$, and $2^{-m} \le 2^{-h}$ because $m = 2^{12h} \ge h$. The sample space is finite, so some outcome has $c_\rho(\mathcal{G})$ at most its expectation. By Step 4 that outcome's $\mathcal{G}$ is a cover of the target family. □

3 Weights for exceptional sets

This section proves Proposition 3.1, an analogue of Theorem 1.1 with only 32 unions and the weaker hypothesis $\mu_q(\mathcal{F}) \ge 1/2$, but with the cover at the lower density q/L, $L = 2^{70}$. It does not by itself prove the theorem; it is used once, in Case 1 of the proof of Theorem 1.1.

Proposition 3.1. *Let $L = 2^{70}$. For every integer $N \geq 1$, every $q \in (0,1)$, and every $\mathcal{F} \subseteq 2^{[N]}$ satisfying $\mu_q(\mathcal{F}) \geq 1/2$, the family $E_{32}(\mathcal{F})$ is (q/L)-small.*

Proof. Plan. We define weights $w(U)$ and prove the global bound (W1) in Step 1 and the local bound (W2) in Step 2. Steps 3–5 build a cover from the heavy sets and from Lemma 2.1 applied to bins of lighter sets, and bound its cost. Step 6 shows that it covers $E_{32}(\mathcal{F})$ by comparing (W2) with a count over bins.

Notation. Identify a subset of $[N]$ with its indicator vector in $\{0,1\}^N$, and write $f = \mathbf{1}_{\mathcal{F}} : \{0,1\}^N \to \{0,1\}$. For $U \subseteq [N]$, $z \in \{0,1\}^U$ and $y \in \{0,1\}^{[N]\setminus U}$, write (z,y) for the point of $\{0,1\}^N$ that agrees with z on U and with y off U. Define

$$b(U) = (1-q)^{|U|} \sum_{z\in\{0,1\}^U} (-1)^{\sum_{i\in U} z_i}\, \mathbb{E}[f(z,Y)], \qquad w(U) = b(U)^2 \geq 0, \tag{3.1}$$

where Y is a random element of $\{0,1\}^{[N]\setminus U}$ with independent Bernoulli-q coordinates. For $U = \varnothing$ the sum has one term and

$$b(\varnothing) = \mathbb{E}[f(Y)] = \mu_q(\mathcal{F}) \geq \tfrac{1}{2}. \tag{3.2}$$

Step 1 (the global weight bound (W1))**.** We show

$$\sum_{U\subseteq[N]} q^{|U|} w(U) \leq 1. \tag{3.3}$$

We use the standard biased Fourier basis [7, Section 8.4], with the normalization made explicit. Let $\xi = (\xi_1, \dots, \xi_N)$ have law μ_q, and for $x \in \{0,1\}^N$ put

$$g_i(x) = \frac{q - x_i}{\sqrt{q(1-q)}}, \qquad \chi_U(x) = \prod_{i\in U} g_i(x) = \frac{\prod_{i\in U}(q-x_i)}{[q(1-q)]^{|U|/2}}, \qquad U \subseteq [N].$$

1a. Orthonormality. $\mathbb{E}[g_i(\xi)] = \big((1-q)q + q(q-1)\big)/\sqrt{q(1-q)} = 0$ and $\mathbb{E}[g_i(\xi)^2] = \big((1-q)q^2 + q(1-q)^2\big)/(q(1-q)) = 1$. By independence of the coordinates,

$$\mathbb{E}[\chi_U(\xi)\chi_V(\xi)] = \prod_{i\in U\cap V} \mathbb{E}[g_i(\xi)^2] \prod_{i\in U\triangle V} \mathbb{E}[g_i(\xi)] = \begin{cases} 1, & U = V, \\ 0, & U \neq V. \end{cases}$$

Since $0 < q < 1$, μ_q has full support, so $\langle g, g'\rangle = \mathbb{E}[g(\xi)g'(\xi)]$ is an inner product on the 2^N-dimensional space of real functions on $\{0,1\}^N$. The 2^N orthonormal functions χ_U therefore form an orthonormal basis. With $\widehat{f}(U) = \mathbb{E}[f(\xi)\chi_U(\xi)]$, Parseval's identity reads

$$\sum_{U\subseteq[N]} \widehat{f}(U)^2 = \mathbb{E}[f(\xi)^2]. \tag{3.4}$$

1b. Relation between $\widehat{f}$ and b. For one coordinate and $z_i \in \{0,1\}$,

$$\mathbb{P}(\xi_i = z_i)\,(q - z_i) = q(1-q)(-1)^{z_i},$$

since $z_i = 0$ gives $(1-q)\cdot q$ and $z_i = 1$ gives $q\cdot(q-1)$. Now condition on the restriction of ξ to U being z. The function χ_U depends only on the coordinates in U, and the remaining coordinates are independent of them with the law of Y. So

$$\widehat{f}(U) = \sum_{z\in\{0,1\}^U} \prod_{i\in U} \frac{\mathbb{P}(\xi_i = z_i)(q - z_i)}{\sqrt{q(1-q)}}\, \mathbb{E}[f(z,Y)] = [q(1-q)]^{|U|/2} \sum_{z\in\{0,1\}^U} (-1)^{\sum_{i\in U} z_i}\, \mathbb{E}[f(z,Y)].$$

Comparing with (3.1),

$$\widehat{f}(U) = \frac{[q(1-q)]^{|U|/2}}{(1-q)^{|U|}}\, b(U) = \left(\frac{q}{1-q}\right)^{|U|/2} b(U), \qquad \text{i.e.} \qquad q^{|U|} w(U) = (1-q)^{|U|}\, \widehat{f}(U)^2.$$

1c. Conclusion. Using 1b, then $(1-q)^{|U|} \le 1$, then (3.4), then $f^2 = f$,

$$\sum_{U \subseteq [N]} q^{|U|} w(U) = \sum_{U \subseteq [N]} (1-q)^{|U|} \widehat{f}(U)^2 \le \sum_{U \subseteq [N]} \widehat{f}(U)^2 = \mathbb{E}[f(\xi)^2] = \mu_q(\mathcal{F}) \le 1.$$

Step 2 (a lower bound on every exceptional set (W2))**.** Fix $S \in E_{32}(\mathcal{F})$. We show

$$\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \ge 2^{-32}. \tag{3.5}$$

2a. A signed measure on arrays. Consider arrays $\mathbf{x} = (x_{ji}) \in \{0,1\}^{32 \times N}$ with 32 rows and N columns. Write $x_j \in \{0,1\}^N$ for row j (a subset of $[N]$) and $x^i \in \{0,1\}^{32}$ for column i. For a column $y \in \{0,1\}^{32}$ let

$$P(y) = \prod_{j=1}^{32} q^{y_j} (1-q)^{1-y_j}, \qquad \widetilde{P}(y) = P(y) - (1-q)^{32} (-1)^{\sum_{j=1}^{32} y_j}.$$

P is the mass function of 32 independent Bernoulli-q bits. $\widetilde{P}$ is a signed correction of P, chosen so that

$$\widetilde{P}(0, \dots, 0) = (1-q)^{32} - (1-q)^{32} = 0, \tag{3.6}$$

and so that the correction term is a product over the 32 rows. Define the signed weight of an array by

$$\nu_S(\mathbf{x}) = \prod_{i \notin S} P(x^i) \prod_{i \in S} \widetilde{P}(x^i).$$

2b. A vanishing sum. We claim

$$\Sigma := \sum_{\mathbf{x} \in \{0,1\}^{32 \times N}} \nu_S(\mathbf{x}) \prod_{j=1}^{32} f(x_j) = 0. \tag{3.7}$$

Indeed, every term is zero. If $\nu_S(\mathbf{x}) \neq 0$, then by (3.6) no column x^i with $i \in S$ is all-zero, so every $i \in S$ lies in some row: $S \subseteq x_1 \cup \cdots \cup x_{32}$. By the definition (1.1) of $E_{32}(\mathcal{F})$, the rows $x_1, \dots, x_{32}$ cannot all belong to $\mathcal{F}$, so $\prod_j f(x_j) = 0$. The argument uses only that the product vanishes wherever $\nu_S \neq 0$; the signs of ν_S play no role.

2c. Expanding Σ. We claim

$$\Sigma = \sum_{U \subseteq S} (-1)^{|U|} b(U)^{32}. \tag{3.8}$$

Write $\widetilde{P} = P + \Delta$ with $\Delta(y) = -(1-q)^{32} (-1)^{\sum_j y_j}$. Expanding the product over $i \in S$, and letting U be the set of columns where Δ is chosen,

$$\nu_S(\mathbf{x}) = \sum_{U \subseteq S} \prod_{i \notin U} P(x^i) \prod_{i \in U} \Delta(x^i).$$

Both factors are products over the rows j:

$$\prod_{i\in U}\Delta(x^i)=(-1)^{|U|}(1-q)^{32|U|}\prod_{j=1}^{32}(-1)^{\sum_{i\in U}x_{ji}},\qquad \prod_{i\notin U}P(x^i)=\prod_{j=1}^{32}\prod_{i\notin U}q^{x_{ji}}(1-q)^{1-x_{ji}}.$$

Distributing $(1-q)^{32|U|}$ as one factor $(1-q)^{|U|}$ per row, the U-term of $\nu_S(\mathbf{x})$ equals

$$(-1)^{|U|}\prod_{j=1}^{32}\beta_U(x_j),\qquad\text{where}\qquad \beta_U(v)=(1-q)^{|U|}(-1)^{\sum_{i\in U}v_i}\prod_{i\notin U}q^{v_i}(1-q)^{1-v_i}\quad(v\in\{0,1\}^N).$$

Summing over arrays is the same as summing independently over the 32 rows. Hence

$$\Sigma=\sum_{U\subseteq S}(-1)^{|U|}\prod_{j=1}^{32}\Big(\sum_{v\in\{0,1\}^N}\beta_U(v)f(v)\Big)=\sum_{U\subseteq S}(-1)^{|U|}\Big(\sum_{v\in\{0,1\}^N}\beta_U(v)f(v)\Big)^{32}.$$

Finally, writing $v=(z,y)$ as in the Notation paragraph, the last product in $\beta_U(v)$ is $\mathbb{P}(Y=y)$. So

$$\sum_v\beta_U(v)f(v)=(1-q)^{|U|}\sum_{z\in\{0,1\}^U}(-1)^{\sum_{i\in U}z_i}\sum_y\mathbb{P}(Y=y)f(z,y)=b(U)$$

by (3.1). This proves (3.8).

Sanity check. Take $N=1$, $\mathcal{F}=\{\varnothing\}$, $S=\{1\}$. Then $S\in E_{32}(\mathcal{F})$, $b(\varnothing)=\mu_q(\mathcal{F})=1-q$, and $b(\{1\})=(1-q)(f(0)-f(1))=1-q$. So the right side of (3.8) is $(1-q)^{32}-(1-q)^{32}=0$, as it should be.

2d. Conclusion. By (3.7) and (3.8), $b(\varnothing)^{32}=-\sum_{\varnothing\neq U\subseteq S}(-1)^{|U|}b(U)^{32}$. With (3.2) and the triangle inequality,

$$2^{-32}\le b(\varnothing)^{32}\le\sum_{\varnothing\neq U\subseteq S}|b(U)|^{32}=\sum_{\varnothing\neq U\subseteq S}w(U)^{16},$$

which is (3.5).

Attribution (not used in the proof). The original manuscript states that, with two rows instead of 32, $\widetilde{P}$ is Li's signed kernel after multiplication by the Bernoulli masses [6, Section 2], and that Park notes the extension of Li's argument to an even number of unions [9, Corollary 3.5, footnote 3]. It also traces the signed-operator approach and its biased Fourier basis to [4, Section 2.2] and [2, Sections 2.2 and 4.1]. Steps 2a–2d above are complete without these references. The comparison with Li's kernel is context only and has not been re-derived here.

Step 3 (heavy sets as generators)**.** Set $\rho=q/L=q\,2^{-70}$, and let

$$\mathcal{G}_0=\{U\subseteq[N]:U\neq\varnothing,\ w(U)\ge 2^{-64|U|}\}.$$

For $U\in\mathcal{G}_0$ with $r=|U|\ge1$,

$$\rho^r=q^r2^{-70r}=2^{-6r}q^r\cdot2^{-64r}\le2^{-6r}q^rw(U)\le2^{-6}q^rw(U).$$

Summing over $\mathcal{G}_0$ and using (3.3),

$$c_\rho(\mathcal{G}_0)\le2^{-6}\sum_{U\in\mathcal{G}_0}q^{|U|}w(U)\le2^{-6}.\tag{3.9}$$

Step 4 (bins of lighter sets and Lemma 2.1)**.** For integers $r\ge1$ and $h\in\mathbb{Z}$ let

$$\mathcal{H}_{r,h}=\{U\subseteq[N]:|U|=r,\ U\notin\mathcal{G}_0,\quad 2^{-(h+1)}<w(U)\le2^{-h}\}.$$

4a. The bins partition the light sets of positive weight. Every nonempty $U \notin \mathcal{G}_0$ with $w(U) > 0$ lies in exactly one bin: the one with $r = |U|$ and h the unique integer satisfying $2^{-(h+1)} < w(U) \le 2^{-h}$. Sets with $w(U) = 0$ lie in no bin.

4b. Every nonempty bin has $h \ge 64r$. A priori h could be any integer, even negative, since (W1) only gives $w(U) \le q^{-|U|}$. But if $U \in \mathcal{H}_{r,h}$, then $U \notin \mathcal{G}_0$ gives $w(U) < 2^{-64r}$ (a weight equal to the threshold is in $\mathcal{G}_0$), while $w(U) > 2^{-(h+1)}$. Hence $h + 1 > 64r$, that is, $h \ge 64r$ for integers.

4c. The hypotheses of Lemma 2.1 hold for each nonempty bin. By (3.3),

$$|\mathcal{H}_{r,h}|\, q^r\, 2^{-(h+1)} < \sum_{U \in \mathcal{H}_{r,h}} q^r w(U) \le 1, \qquad \text{so} \qquad |\mathcal{H}_{r,h}|\, q^r < 2^{h+1}.$$

Since $L \ge 1$ we have $\rho^r = q^r L^{-r} \le q^r$, and therefore

$$|\mathcal{H}_{r,h}|\, \rho^r \le 2^{h+1}.$$

The other hypotheses hold with $X = [N]$: $0 < \rho < 1$, $\mathcal{H}_{r,h} \subseteq \binom{[N]}{r}$, $1 \le r \le 64r \le h$, and $h \ge 64r \ge 64$. So Lemma 2.1 provides a family $\mathcal{G}_{r,h}$ with

$$\mathcal{G}_{r,h} \text{ covers } \{S \subseteq [N] : S \text{ contains at least } 2^{12h} \text{ members of } \mathcal{H}_{r,h}\}, \qquad c_\rho(\mathcal{G}_{r,h}) \le 3 \cdot 2^{-h}. \tag{3.10}$$

Step 5 (the cover and its cost)**.** Only finitely many bins are nonempty, because $[N]$ has finitely many subsets. Let

$$\mathcal{G} = \mathcal{G}_0 \cup \bigcup_{\mathcal{H}_{r,h} \neq \varnothing} \mathcal{G}_{r,h}.$$

The cost of a union is at most the sum of the costs. Every nonempty bin has $r \ge 1$ and $h \ge 64r$ (Step 4b), so by (3.10)

$$\sum_{\mathcal{H}_{r,h} \neq \varnothing} c_\rho(\mathcal{G}_{r,h}) \le 3 \sum_{r \ge 1} \sum_{h \ge 64r} 2^{-h} = 3 \sum_{r \ge 1} 2 \cdot 2^{-64r} = \frac{6}{2^{64} - 1} < \frac{1}{4}. \tag{3.11}$$

With (3.9),

$$c_\rho(\mathcal{G}) \le \frac{1}{64} + \frac{6}{2^{64} - 1} < \frac{1}{2}.$$

Step 6 ($\mathcal{G}$ covers $E_{32}(\mathcal{F})$)**.** Let $S \subseteq [N]$ contain no member of $\mathcal{G}$. We show $S \notin E_{32}(\mathcal{F})$.

Consider the nonempty sets $U \subseteq S$. None of them is in $\mathcal{G}_0$. Those with $w(U) = 0$ contribute nothing to $\sum w(U)^{16}$. By Step 4a each remaining one lies in exactly one nonempty bin $\mathcal{H}_{r,h}$, where $w(U)^{16} \le 2^{-16h}$. Moreover S contains fewer than 2^{12h} members of each nonempty bin; otherwise S would contain a member of $\mathcal{G}_{r,h}$ by (3.10). So the members of one bin contribute *in total* less than $2^{12h} \cdot 2^{-16h} = 2^{-4h}$. Summing over bins, which all have $h \ge 64r$,

$$\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \le \sum_{r \ge 1} \sum_{h \ge 64r} 2^{-4h} = \sum_{r \ge 1} \frac{2^{-256r}}{1 - 2^{-4}} = \frac{16}{15(2^{256} - 1)} < 2^{-32}. \tag{3.12}$$

This contradicts (3.5) if $S \in E_{32}(\mathcal{F})$. Hence every member of $E_{32}(\mathcal{F})$ contains a member of $\mathcal{G}$. Together with Step 5, $\mathcal{G}$ witnesses that $E_{32}(\mathcal{F})$ is (q/L)-small. □

4 Returning to the original density

Proposition 3.1 covers at density q/L. To obtain density p we want to apply it with $q = Lp$. The family $\mathcal{D}$ is only known to be large under μ_p, so we represent a μ_{Lp}-random set as a union of L sets that each have law μ_p.

Proof of Theorem 1.1. Plan. Set $L = 2^{70}$, so $k = 32L = 2^{75}$, and assume $\mu_p(\mathcal{D}) \geq 1 - 1/k$. In each of two cases we construct, on one probability space, random sets $R_1, \ldots, R_L \subseteq [N]$ ("rows"), *each with law exactly* μ_p but dependent on one another. If $Lp < 1$ the rows are disjoint and their union has law μ_{Lp}, and we apply Proposition 3.1. If $Lp \geq 1$ the rows always cover $[N]$, and $E_k(\mathcal{D})$ turns out to be empty.

Step 0 (all rows are in $\mathcal{D}$ with probability at least 31/32)**.** For any random sets $R_1, \ldots, R_L$ each having law μ_p, the union bound gives

$$\mathbb{P}(R_j \in \mathcal{D} \text{ for all } 1 \leq j \leq L) \geq 1 - \sum_{j=1}^{L} \mathbb{P}(R_j \notin \mathcal{D}) = 1 - L(1 - \mu_p(\mathcal{D})) \geq 1 - \frac{L}{k} = \frac{31}{32}. \tag{4.1}$$

No independence between rows is needed.

Case 1: $Lp < 1$.

Step 1.1 (the coupling)**.** For each coordinate $i \in [N]$, independently across coordinates, choose a label $J_i \in \{0, 1, \ldots, L\}$ with

$$\mathbb{P}(J_i = 0) = 1 - Lp, \qquad \mathbb{P}(J_i = j) = p \quad (1 \leq j \leq L).$$

These are nonnegative and sum to 1 because $Lp < 1$. Put $R_j = \{i : J_i = j\}$ and $R = R_1 \cup \cdots \cup R_L$.

- *Each R_j has law μ_p.* For fixed j, the indicators $\mathbf{1}[J_i = j]$, $i \in [N]$, are functions of the independent variables J_i, hence independent, and each equals 1 with probability p.
- *R has law μ_q with $q = Lp \in (0, 1)$.* We have $i \in R$ if and only if $J_i \neq 0$. These events are independent across i and have probability Lp.

Figure 1 illustrates one coordinate with three rows.

	$J_i = 0$	$J_i = 1$	$J_i = 2$	$J_i = 3$
row 1	⓪	❶	⓪	⓪
row 2	⓪	⓪	❶	⓪
row 3	⓪	⓪	⓪	❶
probability	$1 - 3p$	p	p	p

Figure 1: One coordinate of the sparse coupling, illustrated with three rows and $3p < 1$. Each column is one possible value of the label J_i and shows which rows then contain coordinate i. Each row contains it with probability p; the union contains it with probability $3p$. Making these choices independently across coordinates gives the required product laws.

Step 1.2 (the family $\mathcal{F}$ has μ_q-measure at least 1/2)**.** Define

$$\mathcal{F} = \{D_1 \cup \cdots \cup D_L : D_1, \ldots, D_L \in \mathcal{D}\}.$$

On the event in (4.1), $R = R_1 \cup \cdots \cup R_L$ is a union of L members of $\mathcal{D}$, so $R \in \mathcal{F}$. Since R has law μ_q,

$$\mu_q(\mathcal{F}) = \mathbb{P}(R \in \mathcal{F}) \geq \frac{31}{32} \geq \frac{1}{2}.$$

Step 1.3 ($E_{32}(\mathcal{F}) = E_k(\mathcal{D})$)**.** The sets of the form $F_1 \cup \cdots \cup F_{32}$ with $F_j \in \mathcal{F}$ are exactly the sets of the form $D_1 \cup \cdots \cup D_{32L}$ with $D_j \in \mathcal{D}$. In one direction, write each F_j as a union of L members of $\mathcal{D}$ and concatenate. In the other, split a $32L$-tuple into 32 consecutive blocks of length L; the union of each block is in $\mathcal{F}$. By (1.1),

$$E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D}) = E_k(\mathcal{D}).$$

Step 1.4 (conclusion of Case 1)**.** By Step 1.2, Proposition 3.1 applies to $\mathcal{F}$ at density q and shows that $E_{32}(\mathcal{F})$ is (q/L)-small. Since $q/L = p$ and $E_{32}(\mathcal{F}) = E_k(\mathcal{D})$, the family $E_k(\mathcal{D})$ is p-small.

Case 2: $Lp \geq 1$. The argument of Case 1 is unavailable because $q = Lp \geq 1$ lies outside $(0, 1)$, so Proposition 3.1 cannot be applied at density Lp (and for $Lp > 1$ the label probability $1 - Lp$ would be negative). A different coupling is used.

Step 2.1 (the coupling)**.** Set

$$\tau = \frac{p - 1/L}{1 - 1/L}.$$

Then $\tau \geq 0$ because $p \geq 1/L$, and $\tau < 1$ because $p < 1$. For each coordinate i, independently across coordinates: choose a "mandatory" row K_i uniformly in $\{1, \ldots, L\}$; put i into R_{K_i}; and, conditionally on K_i, put i into each other row independently with probability τ. For a fixed row j,

$$\mathbb{P}(i \in R_j) = \mathbb{P}(K_i = j) + \mathbb{P}(K_i \neq j)\,\tau = \frac{1}{L} + \left(1 - \frac{1}{L}\right)\tau = \frac{1}{L} + p - \frac{1}{L} = p,$$

and the events $\{i \in R_j\}$, $i \in [N]$, are independent because the choices at different coordinates are. So each R_j has law μ_p. Moreover $R_1 \cup \cdots \cup R_L = [N]$ in every outcome, since $i \in R_{K_i}$.

Step 2.2 (conclusion of Case 2)**.** By (4.1) the event that all L rows lie in $\mathcal{D}$ has positive probability, so it occurs for some outcome. This gives $D_1, \ldots, D_L \in \mathcal{D}$ with $D_1 \cup \cdots \cup D_L = [N]$. Repeating this tuple 32 times gives exactly $32L = k$ members of $\mathcal{D}$ whose union is $[N]$, which contains every $S \subseteq [N]$. By (1.1), $E_k(\mathcal{D}) = \varnothing$. The empty family covers it at cost $0 \leq 1/2$. The boundary case $Lp = 1$ is included, with $\tau = 0$. □

5 Consequences and context

Nothing in this section is used in the proof of Theorem 1.1. Section 5.1 is a self-contained consequence of the theorem. Section 5.2 is a separate result whose proof depends on three cited papers. Section 5.3 lists related work as described in the original manuscript.

5.1 Positive selector processes

The link between discrete convexity and positive selector processes is explained in [11, Section 7] and [12, Proposition 13.3.3]. Concretely, let $\mathcal{T} \subseteq [0, \infty)^N$ be nonempty and define

$$\phi(S) = \sup_{\theta \in \mathcal{T}} \sum_{i \in S} \theta_i \quad (S \subseteq [N]), \qquad M = \mathbb{E}_{\mu_p} \phi,$$

and suppose $0 < M < \infty$. Then $\phi(S) < \infty$ for *every* S, because $M = \sum_S \mu_p(\{S\})\phi(S)$ and every $\mu_p(\{S\})$ is positive.

Claim. *With $k = 2^{75}$, the family $\{S : \phi(S) \geq k^2 M\}$ is p-small.*

Proof. We apply Theorem 1.1 to one particular family, which plays the role of $\mathcal{D}$ there.

(1) Let $\mathcal{D} = \{D \subseteq [N] : \phi(D) < kM\}$. Since $\phi \geq 0$ and $M > 0$, Markov's inequality gives $\mu_p\{\phi \geq kM\} \leq 1/k$, so $\mu_p(\mathcal{D}) \geq 1 - 1/k$.

(2) ϕ is monotone: if $A \subseteq B$ then $\phi(A) \leq \phi(B)$, because all $\theta_i \geq 0$. And ϕ is subadditive under unions: for every $\theta \in \mathcal{T}$, $\sum_{i \in A \cup B} \theta_i \leq \sum_{i \in A} \theta_i + \sum_{i \in B} \theta_i \leq \phi(A) + \phi(B)$, and taking the supremum over θ gives $\phi(A \cup B) \leq \phi(A) + \phi(B)$.

(3) If $S \subseteq D_1 \cup \cdots \cup D_k$ with all $D_j \in \mathcal{D}$, then by (2) $\phi(S) \leq \sum_{j=1}^{k} \phi(D_j) < k \cdot kM$. Hence $\{S : \phi(S) \geq k^2 M\} \subseteq E_k(\mathcal{D})$.

(4) By (1), Theorem 1.1 gives a cover of $E_k(\mathcal{D})$ of cost at most $1/2$ at density p. By (3) it also covers $\{S : \phi(S) \geq k^2 M\}$. □

The original manuscript notes that this recovers the positive-selector conclusion proved by Park and Pham [10, Theorem 1.2], a statement of the same shape (the set where ϕ exceeds a universal constant multiple of M is p-small), here with the constant $k^2 = 2^{150}$. As for what the claim adds: Markov's inequality alone bounds only the probability, $\mu_p\{\phi \geq k^2 M\} \leq k^{-2}$. The claim supplies a structural certificate of a different kind, namely an explicit list $\mathcal{G}$ of sets with $\sum_{I \in \mathcal{G}} p^{|I|} \leq 1/2$ such that every S with $\phi(S) \geq k^2 M$ contains a member of $\mathcal{G}$.

5.2 Two unions at constant density loss

Theorem 1.1 keeps the density p but uses 2^{75} unions and needs $\mu_p(\mathcal{D}) \geq 1 - 2^{-75}$. Corollary 5.1 below, which the original manuscript states in its introduction, sits at the other end of the tradeoff: two unions, hypothesis $\mu_p(\mathcal{D}) > 1/2$, and a constant factor lost in the density. Its proof does not use Sections 2–4; it combines external results.

Fractional smallness. For $\sigma \in (0, 1)$, a family $\mathcal{A}$ has a *fractional cover of cost at most* $1/2$ *at density* σ if there are coefficients $a_I \geq 0$ ($I \subseteq [N]$) with

$$\sum_{I \subseteq S} a_I \geq 1 \quad (S \in \mathcal{A}), \qquad \sum_{I \subseteq [N]} \sigma^{|I|} a_I \leq \frac{1}{2}.$$

A cover in the sense of Section 1 is the special case $a_I \in \{0, 1\}$; it is called *integral* below. Talagrand introduced this relaxation alongside integral smallness [11, Sections 6 and 7].

The cited inputs, in the form the proof uses them. Both are applied to the family $\mathcal{A} = E_2(\mathcal{D})$, after it has been shown to be nonempty, proper and increasing. The original manuscript does not say which of these three properties each citation requires.

(I1) *Li's result [6, Corollary 3.1] in Park's fractional-cover formulation [9, Theorem 3.4].* As invoked: if $\mu_p(\mathcal{D}) > 1/2$ and $\mathcal{A} = E_2(\mathcal{D})$ is a nonempty proper increasing family, then $\mathcal{A}$ has a fractional cover of cost at most $1/2$ at density $\sigma = p/2$.

(I2) *The companion theorem [8, Theorem 1.1].* The original manuscript calls it both the "companion constant-factor rounding theorem" and the "companion expectation-threshold theorem"; both refer to this one citation. As invoked: if the nonempty proper increasing family $\mathcal{A}$ has a fractional cover of cost at most $1/2$ at density σ with all coefficients in $[0, 1]$, then the same family $\mathcal{A}$ has an integral cover of cost at most $1/2$ at density $\sigma/(25 \cdot 512^4)$. The full strength

of the hypothesis $\mu_p(\mathcal{D}) > 1/2$ is not used here; it enters only through (I1). What (I2) does rely on is the weaker consequence $\mathcal{D} \neq \varnothing$ of that hypothesis, which gives the properness of $\mathcal{A}$ in step (2) of the proof below.

Here "proper" is read as "not all of $2^{[N]}$". For an increasing family this is the same as not containing $\varnothing$, which is what the proof checks.

Corollary 5.1 (Two unions at constant density loss)**.** *There is a universal constant $C \geq 1$ such that, for every integer $N \geq 1$, every $p \in (0,1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) > \frac{1}{2} \quad \Longrightarrow \quad E_2(\mathcal{D}) \text{ is } (p/C)\text{-small.}$$

Proof. (1) If $E_2(\mathcal{D}) = \varnothing$, the empty cover suffices.

(2) Otherwise $E_2(\mathcal{D})$ is nonempty. It is increasing by the second bullet after (1.1). It is proper: $\mu_p(\mathcal{D}) > 0$ gives $\mathcal{D} \neq \varnothing$, so $\varnothing \notin E_2(\mathcal{D})$ by the first bullet there.

(3) By (I1), $E_2(\mathcal{D})$ has a fractional cover (a_I) of cost at most $1/2$ at density $\sigma = p/2$.

(4) Replace each a_I by $a'_I = \min\{a_I, 1\}$. The cost does not increase. Coverage is preserved: for $S \in E_2(\mathcal{D})$, if some $I \subseteq S$ has $a_I \geq 1$ then $\sum_{I \subseteq S} a'_I \geq 1$, and otherwise the sum is unchanged.

(5) By (I2), $E_2(\mathcal{D})$ has an integral cover of cost at most $1/2$ at density $\sigma/(25 \cdot 512^4) = p/(50 \cdot 512^4)$. Thus $C = 50 \cdot 512^4$ suffices. □

Unverified step. Steps (3) and (5) rest on (I1) and (I2). The original manuscript invokes [6, Corollary 3.1], [9, Theorem 3.4] and [8, Theorem 1.1] by number without reproducing their statements, so I could not check that their hypotheses are met by $E_2(\mathcal{D})$ under $\mu_p(\mathcal{D}) > 1/2$, why the density in (I1) is $p/2$, or the constant $25 \cdot 512^4$ in (I2). Input (I2) is a strong claim (rounding a fractional cover to an integral one with only a constant-factor loss in density), and it alone carries the improvement over Park's result discussed below. Steps (1), (2), (4) and the arithmetic in (5) are verified. The corollary is not used for Theorem 1.1.

Which citation is which, and the comparison with Park's result. The original manuscript cites four different statements from [6] and [9]. Its descriptions are as follows. [6, Theorem 1.9] resolves the fractional discrete-convexity problem through a quantitative obstruction to probability measures whose containment probabilities are bounded by $p^{|I|}$; it is cited as context. [6, Corollary 3.1] and [9, Theorem 3.4] are the forms invoked in (I1). [9, Corollary 3.5] is Park's own integral result: combining fractional information with rounding, Park obtains an integral cover of $E_2(\mathcal{D})$ at density

$$\frac{p}{C \max\{1, \log\log(1/p)\}}$$

whenever $\mu_p(\mathcal{D}) > 1/2$, for a universal C. Corollary 5.1 has the same hypothesis and replaces this density by p/C. Nothing proved in the present document explains this improvement. The logic is only this: the fractional cover in step (3) is already at density $p/2$, with no $\log\log$ factor, so any such factor in an integral result can only come from the rounding step; and the manuscript replaces the rounding by (I2), which is asserted to lose only a constant factor (the cited preprint is titled "Integral and fractional expectation thresholds are equivalent"). Whether (I2) holds as invoked is exactly what the box above flags.

5.3 Related developments

The following descriptions are those of the original manuscript.

Origin of the name. The name reflects the analogy with Talagrand's Gaussian convexity question [11, Conjecture 2.1]: can a bounded number of sums of a compact balanced set of sufficiently large Gaussian measure contain a convex set of Gaussian measure at least $1/2$, uniformly in the dimension? Here balanced means closed under multiplication by scalars of absolute value at most one. In the discrete problem, unions replace sums, and containment covers describe the exceptional sets.

Other results. Hua, Song, and Tudose resolve Talagrand's Gaussian convexity problem and derive a discrete conclusion with three unions and covering density p^C, for a universal constant C, when the starting family is sufficiently likely [5, Corollaries 1.3–1.4]. Ascoli, He, Park, and Talagrand give a threshold reformulation of discrete convexity and prove cases for several graph-containment properties [1, Proposition 1.5 and Theorems 1.10–1.13].

Antecedents of the method. The signed product of Step 2 of Proposition 3.1 has antecedents in the signed product operators and biased Fourier analysis of Friedgut [4, Section 2.2] and Ellis, Filmus, and Friedgut [2, Sections 2.2 and 4.1]. Its two-row version is attributed to Li [6, Section 2]. Random unions also enter the covering constructions of Fischer and Person [3, Sections 1.1 and 2].

6 Verification checklist

A sceptical reader should check these load-bearing steps first.

1. **Extension inequality** (2.3) (Lemma 2.1, Step 2). Check that a nonempty group $\{e \in \mathcal{H}' : e \cap B = I\}$ forces $d(I) \le T$ (because I lies in a regular edge, hence is not in $\mathcal{C}_0$), and that the number of groups is at most $\sum_{i \le r} \binom{|B|}{i}$. This inequality drives both the pair cost (2.4) and the recursion (2.5).

2. **Survival probability and arithmetic** (Lemma 2.1, Steps 5b–5c). The m^2 pairs of a tuple of distinct edges are distinct sampling indices, diagonal included, giving $(1-\pi)^{m^2}$. Then check $4h + 13rh + 1 \le 2^{4h} = \pi m$, which uses $r \le h$, $h \le 2^h$, and $18 \le 2^{2h}$ (true for $h \ge 3$).

3. **$\mathcal{G}$ is a cover for every outcome** (Lemma 2.1, Step 4). Every edge inside a set avoiding $\mathcal{C}_0$ is regular, and the dichotomy "some pair sampled / no pair sampled" is exhaustive.

4. **Normalization $\widehat{f}(U) = (q/(1-q))^{|U|/2} b(U)$ and (W1)** (Proposition 3.1, Step 1b–1c).

5. **Signed identity** (3.8) (Proposition 3.1, Step 2). Two points: $\widetilde{P}(0,\dots,0) = 0$ makes every term of Σ vanish (2b), and the U-term factors over the 32 rows with each row giving exactly $b(U)$ (2c).

6. **Bin bookkeeping** (Proposition 3.1, Steps 4–6). Nonempty bins have $h \ge 64r$; $|\mathcal{H}_{r,h}|\rho^r \le 2^{h+1}$; the two geometric sums (3.11) and (3.12); and the comparison $16/(15(2^{256}-1)) < 2^{-32} \le$ the bound (3.5).

7. **The couplings** (Section 4). In Case 1 each row has law exactly μ_p, the union has law exactly μ_{Lp}, and $E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D})$ (Steps 1.1–1.3). In Case 2 each row has law μ_p and the rows always cover $[N]$ (Step 2.1).

8. **Scope.** Corollary 5.1 depends on the external inputs (I1) and (I2), which are not verified here (see the box in Section 5.2). Theorem 1.1 does not depend on it.

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